

Amazon Sales Data



SQL Project Presentation

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Overview Of Amazon Sales Data

- The data consists of sales record of three cities/branch in Myanmar which are Naypyitaw, Yangon, Mandalay which took place in first quarter of year 2019 . The data consists of 1000 rows and 17 columns.

Objective of Project

- The major aim of this project is to gain insight into the sales data of Amazon to understand the different factors that affect sales of the different branches

Preview of Amazon Sales Data

Column	Description	Data Type
Invoice Id	Invoice of the sales made	Varchar(30)
Branch	Branch at which sales were made	Varchar(5)
City	The location of the branch	Varchar(30)
Customer Type	The type of the customer	Varchar(30)
Gender	Gender of the customer making purchase	Varchar(10)
Product Line	Product line of the product sold	Varchar(100)
Unit Price	The price of each product	Decimal(10,2)
Quantity	The amount of the product sold	Int
VAT	The amount of tax on the purchase	Float
Total	The total cost of the purchase	Decimal(10,2)
Date	The date on which the purchase was made	Date
Time	The time at which the purchase was made	Time
Payment Method	The total amount paid	Varchar(15)
Cogs	Cost Of Goods sold	Decimal(10,2)
Gross Margin Percentage	Gross margin percentage	Float
Gross Income	Gross Income	Decimal(10,2)
Rating	Rating	Decimal(3,1)

Data Wrangling

Step [1]: Created a database named Amazon in MySQL.



Step [2]: Importing data in the form of a demo table named Amazon using table data import wizard.



	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	Total	Date	Time	Payment	cogs	gross margin percentage	gross income	Rating
▶	750-67-8428	A	Yangon	Member	Female	Health and beauty	74.69	7	26.1415	548.9715	2019-01-05	13:08:00	Ewallet	522.83	4.761904762	26.1415	9.1
	226-31-3081	C	Naypyitaw	Normal	Female	Electronic accessories	15.28	5	3.82	80.22	2019-03-08	10:29:00	Cash	5.4	4.761904762	3.82	9.6
	631-41-3108	A	Yangon	Normal	Male	Home and lifestyle	46.33	7	16.2155	340.5255	2019-03-03	13:23:00	Credit card	324.31	4.761904762	16.2155	7.4
	123-19-1176	A	Yangon	Member	Male	Health and beauty	58.22	8	23.288	489.048	2019-01-27	20:33:00	Ewallet	465.76	4.761904762	23.288	8.4
	373-73-7910	A	Yangon	Normal	Male	Sports and travel	86.31	7	30.2085	634.3785	2019-02-08	10:37:00	Ewallet	604.17	4.761904762	30.2085	5.3
	699-14-3026	C	Naypyitaw	Normal	Male	Electronic accessories	85.39	7	29.8865	627.6165	2019-03-25	18:30:00	Ewallet	597.73	4.761904762	29.8865	4.1
	355-53-5943	A	Yangon	Member	Female	Electronic accessories	68.84	6	20.652	433.692	2019-02-25	14:36:00	Ewallet	413.04	4.761904762	20.652	5.8

Step [3]: Checking null values and datatypes of columns of demo amazon table.

Note: as observe the datatype are incorrect and column names contain space which is syntactically incorrect, also table has no null values. This correction is done in EDA.

7 • describe amazon;

Field	Type	Null	Key	Default	Extra
Invoice ID	text	YES		NULL	
Branch	text	YES		NULL	
City	text	YES		NULL	
Customer type	text	YES		NULL	
Gender	text	YES		NULL	
Product line	text	YES		NULL	
Unit price	double	YES		NULL	
Quantity	int	YES		NULL	
Tax 5%	double	YES		NULL	
Total	double	YES		NULL	
Date	text	YES		NULL	
Time	text	YES		NULL	
Payment	text	YES		NULL	
cogs	double	YES		NULL	
gross margin ...	double	YES		NULL	
gross income	double	YES		NULL	
Rating	double	YES		NULL	

9 • select count(*) as count_of_null_values from amazon
10 where null;

count_of_null_values
0

Feature Engineering

In this step we are creating new columns named **timeofday**, **dayname**, **monthname** by extracting values from date and time column. This will help us to analyse and answer sales based on time-of-day (Morning, Afternoon, Evening), day-of-week (Sunday to Saturday) and month (Jan-March).

```
81 • select 'invoice id', date, time, time_of_day, day_name, month_name from amazon
82   limit 5
```

	invoice id	date	time	time_of_day	day_name	month_name
▶	750-67-8428	2019-01-05	13:08:00	Afternoon	Saturday	January
	226-31-3081	2019-03-08	10:29:00	Morning	Friday	March
	631-41-3108	2019-03-03	13:23:00	Afternoon	Sunday	March
	123-19-1176	2019-01-27	20:33:00	Evening	Sunday	January
	373-73-7910	2019-02-08	10:37:00	Morning	Friday	February

```
46 • alter table amazon
47   add time_of_day varchar(15) not null;
48
49 • update amazon set time_of_day =
50   case
51     when hour(time) between 06 and 11 then 'Morning'
52     when hour(time) between 12 and 17 then 'Afternoon'
53     else 'Evening'
54   end;
55
56 • alter table amazon
57   add day_name varchar(10) not null;
58
59 • update amazon set day_name =
60   (select dayname(date));
61
62 • alter table amazon
63   add month_name varchar(10) not null;
64
65 • update amazon set month_name =
66   (select monthname(date));
```


Exploratory Data Analysis

Step [1]: Creating new table named **Amazon Sales** by adding correct column names, datatypes, constraints while copying values from demo table Amazon.

```
17 • create table amazon_sales
18   (invoice_id varchar(30) primary key not null,
19    branch varchar(5) not null,
20    city varchar(30) not null,
21    customer_type varchar(30) not null,
22    gender varchar(10) not null,
23    product_line varchar(100) not null,
24    unit_price decimal(10,2) not null,
25    quantity int not null,
26    vat float not null,
27    total decimal(10,2) not null,
28    date date not null,
29    time time not null,
30    payment_method varchar(20) not null,
31    cogs decimal(10,2) not null,
32    gross_margin_percentage float not null,
33    gross_income decimal(10,2) not null,
34    rating decimal(3,1) not null,
35    time_of_day varchar(15) not null,
36    day_name varchar(10) not null,
37    month_name varchar(10) not null);
38
```

42 • describe amazon_sales;

Result Grid					
Filter Rows:					
Export:					
	Field	Type	Null	Key	Default
▶	invoice_id	varchar(30)	NO	PRI	NULL
	branch	varchar(5)	NO		NULL
	city	varchar(30)	NO		NULL
	customer_type	varchar(30)	NO		NULL
	gender	varchar(10)	NO		NULL
	product_line	varchar(100)	NO		NULL
	unit_price	decimal(10,2)	NO		NULL
	quantity	int	NO		NULL
	vat	float	NO		NULL
	total	decimal(10,2)	NO		NULL
	date	date	NO		NULL
	time	time	NO		NULL
	payment_met...	varchar(20)	NO		NULL
	cogs	decimal(10,2)	NO		NULL
	gross_margin...	float	NO		NULL
	gross_income	decimal(10,2)	NO		NULL

Step [2]: Checking size of table, count of null values, unique values in columns.

```
71 • select count(*) as total_columns from information_schema.columns
72   where table_name = 'amazon_sales';
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	total_columns			
▶	20			

```
75 • select count(*) as total_rows from amazon_sales;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	total_rows			
▶	1000			

```
77 • select count(*) as null_values from amazon_sales where null;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	null_values			
▶	0			

```
86 • select * from unique_values;
```

Result Grid																					Filter Rows:	Export:	Wrap Cell Contents:
	invoice_id	branch	city	customertype	gender	product_line	unit_price	quantity	vat	total	date	time	payment_method	cogs	gross_margin_percentage	gross_income	rating	time_of_day	day_name	month_name			
▶	1000	3	3	2	2	6	943	10	990	990	89	506	3	990	1	873	61	3	7	3			

Step [3]: Checking the unique values in each categorical column. There are 10 categorical columns [**invoice_id**, **branch**, **city**, **customer_type**, **gender**, **product_line**, **payment_method**, **time_of_day**, **day_name**, **month_name**]

	branch
	A
	C
▶	B

	city
▶	Yangon
	Naypyitaw
	Mandalay

	time_of_day
	Evening
▶	Afternoon
	Morning

	month_name
▶	March
	January
	February

	payment_method
▶	Credit card
	Ewallet
	Cash

	gender
▶	Male
	Female

	customer_type
▶	Normal
	Member

	day_name
▶	Wednesday
	Thursday
	Tuesday
	Friday
	Monday
	Saturday
	Sunday

	product_line
▶	Food and beverages
	Health and beauty
	Sports and travel
	Fashion accessories
	Home and lifestyle
	Electronic accessories

Answering Business Questions

Q.1] What is the count of distinct cities in the dataset?

```
5 • select count(distinct(city)) from amazon_sales;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	count(distinct(city))			
▶	3			

Q.2] For each branch, what is corresponding city?

```
5 • select distinct city, branch from amazon_sales;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	city	branch		
▶	Yangon	A		
	Naypyitaw	C		
	Mandalay	B		

Q.3] What is the count of distinct product lines in the dataset?

```
8 • select count(distinct(product_line)) from amazon_sales;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	count(distinct(product_line))			
▶	6			

Q.4] Which payment method occurs most frequently?

```
11 • select payment_method, count(*) as occurrence from amazon_sales
12     group by payment_method
13     order by occurrence desc;
```

	payment_method	occurrence
▶	Ewallet	345
	Cash	344
	Credit card	311

Q.5] Which product line has the highest sales?

```
16 • select product_line, sum(quantity) as total_sales from amazon_sales
17     group by product_line
18     order by total_sales desc;
```

	product_line	total_sales
▶	Electronic accessories	971
	Food and beverages	952
	Sports and travel	920
	Home and lifestyle	911
	Fashion accessories	902
	Health and beauty	854

Q.6] How much revenue is generated each month?

```
21 • select month_name, sum(total) as monthly_revenue$ from amazon_sales
22     group by month_name
23     order by monthly_revenue$ desc;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	month_name	monthly_revenue\$			
▶	January	116292.11			
	March	109455.74			
	February	97219.58			

Q.7] Which product line generated highest revenue?

```
31 • select product_line, sum(total) as total_revenue$ from amazon_sales
32     group by product_line
33     order by total_revenue$ desc;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	product_line	total_revenue\$			
▶	Food and beverages	56144.96			
	Sports and travel	55123.00			
	Electronic accessories	54337.64			
	Fashion accessories	54306.03			
	Home and lifestyle	53861.96			
	Health and beauty	49193.84			

Q.8] In which month cost of goods sold reach its peak?

```
26 • select month_name, sum(cogs) as cost_of_goods_sold from amazon_sales
27     group by month_name
28     order by cost_of_goods_sold desc;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	month_name	cost_of_goods_sold			
▶	January	110754.16			
	March	104243.34			
	February	92589.88			

Q.9] Which city has the highest revenue recorded?

```
36 • select city, sum(total) as revenue$ from amazon_sales
37     group by city
38     order by revenue$ desc;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	city	revenue\$			
▶	Naypyitaw	110568.86			
	Yangon	106200.57			
	Mandalay	106198.00			

Q.10] Which product line incurred the highest value added tax?

```
41 • select product_line, max(vat) highest_vat from amazon_sales
42     group by product_line
43     order by highest_vat desc;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	product_line	highest_vat			
▶	Fashion accessories	49.65			
	Food and beverages	49.26			
	Home and lifestyle	48.75			
	Sports and travel	47.72			
	Health and beauty	45.25			
	Electronic accessories	44.8785			

Q.11] Which customer type occurs most frequently?

```
102 • select customer_type, count(*) as count from amazon_sales
103     group by customer_type
104     order by count desc;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	customer_type	count			
▶	Member	501			
	Normal	499			

Q.12] For each product line, add a column indicating "Good" if its sales are above average, otherwise "Bad."

```
46 • select product_line, sum(total) as revenue,  
47     case  
48         when sum(total) > (select sum(total)/count(distinct(product_line)) from amazon_sales) then 'Good'  
49         else 'Bad'  
50     end performance  
51     from amazon_sales  
52     group by product_line;
```

Result Grid			
Filter Rows:		Export:	Wrap Cell Content:
product_line	revenue	performance	
Food and beverages	56144.96	Good	
Health and beauty	49193.84	Bad	
Sports and travel	55123.00	Good	
Fashion accessories	54306.03	Good	
Home and lifestyle	53861.96	Good	
Electronic accessories	54337.64	Good	

Q.13] Which branch exceeded the average number of product sold?

```
55 • select branch, sum(quantity) as product_sold from amazon_sales  
56     group by branch  
57     having product_sold > (select sum(quantity)/count(distinct branch) as avg_quantity from amazon_sales);
```

Result Grid	
Filter Rows:	
branch	product_sold

Q.14] Which product line is most frequently associated with each gender?

```
60 • with new as
61   (select gender, product_line, count(*) as count from amazon_sales
62    group by gender, product_line),
63
64   max_count as
65   (select max(count) from new group by gender)
66
67   select * from new
68   where count in (select * from max_count) limit 2;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
gender	product_line	count	
Male	Health and beauty	88	
Female	Fashion accessories	96	

Q.15] What is the count of distinct customer types in the dataset?

```
96 • select count(distinct(customer_type)) as count_distinct_customer_type from amazon_sales;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
count_distinct_customer_type			
2			

Q.16] Calculate the average rating for each product line.

```
71 • select product_line, avg(rating) as avg_rating from amazon_sales
72     group by product_line;
```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 		
	product_line	avg_rating
▶	Food and beverages	7.11322
	Health and beauty	7.00329
	Sports and travel	6.91627
	Fashion accessories	7.02921
	Home and lifestyle	6.83750
	Electronic accessories	6.92471

Q.17] Identify the customer type contributing the highest revenue.

```
81 • select customer_type, sum(total) as revenue from amazon_sales
82     group by customer_type
83     order by revenue desc;
```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 		
	customer_type	revenue
▶	Member	164223.81
	Normal	158743.62

Q.18] Count the sales occurrences for each time of day on every weekday.

```
75 • select day_name, time_of_day, count(*) sales from amazon_sales
76     group by day_name, time_of_day
77     order by field(day_name, 'Sunday', 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday'),
78            field(time_of_day, 'Morning', 'Afternoon', 'Evening');
```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 			
	day_name	time_of_day	sales
▶	Sunday	Morning	22
	Sunday	Afternoon	70
	Sunday	Evening	41
	Monday	Morning	21
	Monday	Afternoon	75
	Monday	Evening	29
	Tuesday	Morning	36
	Tuesday	Afternoon	71
	Tuesday	Evening	51
	Wednesday	Morning	22
	Wednesday	Afternoon	81
	Wednesday	Evening	40
	Thursday	Morning	33
	Thursday	Afternoon	76
	Thursday	Evening	29
	Friday	Morning	29
	Friday	Afternoon	74
	Friday	Evening	36
	Saturday	Morning	28
	Saturday	Afternoon	81
	Saturday	Evening	55

Q.19] Determine city with highest VAT percentage.

```
86 • select city, max(vat) as vat_percentage from amazon_sales
87     group by city
88     order by vat_percentage desc;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	city	vat_percentage			
▶	Naypyitaw	49.65			
	Yangon	49.49			
	Mandalay	48.69			

Q.20] Identify the customer type with the highest VAT payments.

```
91 • select customer_type, max(vat) as vat_percentage from amazon_sales
92     group by customer_type
93     order by vat_percentage desc;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	customer_type	vat_percentage			
▶	Member	49.65			
	Normal	49.49			

Q.21] What is the count of distinct payment methods in the dataset?

```
99 • select count(distinct(payment_method)) as count_distinct_payment from amazon_sales;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
count_distinct_payment			
▶ 3			

Q.22] Examine distribution of gender within each branch.

```
117 • select branch, gender, count(*) as count from amazon_sales
118      group by branch, gender
119      order by branch, gender;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
branch	gender	count	
▶ A	Female	161	
A	Male	179	
B	Female	162	
B	Male	170	
C	Female	178	
C	Male	150	

Q.23] Determine predominant gender among customer.

```
112 • select gender, count(*) as count from amazon_sales
113     group by gender
114     order by count desc;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
gender	count			
Female	501			
Male	499			

Q.24] Identify the day of the week with the highest average ratings.

```
134 • select day_name, avg(rating) as avg_rating from amazon_sales
135     group by day_name
136     order by avg_rating desc;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
day_name	avg_rating			
Monday	7.15360			
Friday	7.07626			
Sunday	7.01128			
Tuesday	7.00316			
Saturday	6.90183			
Thursday	6.88986			
Wednesday	6.80559			

Q.25] Identify the time of day when customer provide most ratings.

```
122 • select time_of_day, count(rating) as rating_count from amazon_sales
123      group by time_of_day
124      order by rating_count desc;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	time_of_day	rating_count			
▶	Afternoon	528			
	Evening	281			
	Morning	191			

Q.26] Determine the time of day with the highest customer ratings for each branch.

```
127 • select branch, time_of_day, max(rating) highest_rating from amazon_sales
128      group by branch, time_of_day
129      having highest_rating = (select max(x.max) from (select branch, time_of_day, max(rating) max from amazon_sales
130      group by branch, time_of_day) as x where x.branch= amazon_sales.branch)
131      order by branch;
```

Result Grid				Filter Rows:	Export:	Wrap Cell Content:
	branch	time_of_day	highest_rating			
▶	A	Afternoon	10.0			
	B	Afternoon	10.0			
	B	Evening	10.0			
	B	Morning	10.0			
	C	Afternoon	10.0			

Q.27]. Determine the day of the week with the highest average ratings for each branch.

```
139 • with avg_rating as
140     (select branch, day_name, avg(rating) avg_rat from amazon_sales
141      group by branch, day_name),
142
143     max_rating as
144     (select max(avg_rat) from avg_rating group by branch)
145
146     select branch, day_name, avg_rat as highest_avg_rat from avg_rating where avg_rat in (select * from max_rating);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	branch	day_name	highest_avg_rat
▶	A	Friday	7.31200
	B	Monday	7.33590
	C	Friday	7.27895

Key Findings

Product Analysis:

- Highest Sales Product Line: **Electronic Accessories (Units Sold:971)**
- Highest Revenue Product Line: **Food and Beverages (\$ 56144.96)**
- Lowest Sales Product Line: **Health and Beauty (Unit Sold: 854)**
- Lowest Revenue Product Line: **Health and Beauty (\$ 49193.84)**

Sales Analysis:

- Month With Highest Revenue: **January (\$ 116292.11)**
- City & Branch With Highest Revenue: **Naypyitaw[C] (\$ 110568.86)**
- Month With Lowest Revenue: **February (\$ 97219.58)**
- City & Branch With Lowest Revenue: **Mandalay[B] (\$ 106198.00)**
- Peak Sales Time Of Day: **Afternoon**

- Peak Sales Day Of Week: **Saturday**

Customer Analysis:

- Most Predominant Gender: **Female**
- Most Predominant Customer Type: **Member**
- Highest Revenue Gender: **Female (\$ 167883.26)**
- Highest Revenue Customer Type: **Member (\$ 164223.81)**
- Most Popular Product Line (Male): **Health and Beauty**
- Most Popular Product Line (Female): **Fashion Accessories**
- Distribution Of Members Based On Gender: **Male(240) Female(261)**
- Sales Male: **2641 units**
- Sales Female: **2869 units**



THANK YOU